Predicting Customer Churn in Retail Banking

Finding customers who are about to leave, before they go

MEQDAD MUHANA | General Assembly - DSB FT2

2026

The Churn Challenge in Retail Banking

Savings accounts fund the bank. Customers often drain balances and go dormant without formally closing, so the bank discovers churn after the money has gone.

The Scale

28,382 customers
21 columns
18.53% churn rate

The Aim

Classify every customer as high risk or low risk so the team contacts only those likely to leave.

Success Metrics

- Recall ≥ 0.70
- ROC-AUC ≥ 0.80

The Data Landscape

Source: Kaggle, Bank Churn Data

DATA QUALITY

- NO DUPLICATED ROWS
- 3,871 MISSING VALUES
- HIGH SKEW (5.9M MAX)

Customer
6 fields: age, gender, occupation, dependents...

Location
City and Branch location codes.

Balance
6 fields across current/prior month and two quarters.

Activity
4 transaction fields: money in and out (monthly).

Recency
Date of last transaction recorded for each customer.

Target
Binary: 1 = Churned
0 = Stayed.

Churn Is a Two-Month Drain

Churn is not sudden. It is a visible, predictable depletion of funds over approximately 60 days.

Actionable Insight: The bank must act when withdrawal spikes occur, rather than waiting for the balance to hit zero.

Period	Stayed	Churned
`average_monthly_balance_prevQ2` (2 quarters ago)	3,383	3,206
`average_monthly_balance_prevQ` (last quarter)	3,508	3,721
`previous_month_debit` (money out, last month)	43	838
`current_month_debit` (money out, this month)	22	1,120
`current_balance` (now)	3,643	1,541

The Data Was Hiding Its Own Signal

THREE PROBLEMS THE DATA HID

Found	Detail
3,223 Nulls	Dates stored as text "NaT"
806 Ages	Minimum age was 1 year old
10 Deps	Values up to 52

1

Skew Correction

Skew up to **143** was breaking raw correlations. Log transformation restored the signal.

2

Correlation Boost

Feature	Raw	Log
Ex. Current Debit	0.048	**0.236**

3

Categorical Power

Chi-Square: All categorical features significant .

Model the Change, Not Just the Level

The strategy converts raw banking snapshots into features that capture the dynamics of churn.

Engineering

5 new features as: balance_drop_pct, debit_to_balance.

Selection

Dropped of highly correlated balance columns; 24 final features.

Preprocessing

Yeo-Johnson for skew, Frequency encoding for high-cardinality locations.

Imbalance

Compared SMOTE vs. **class_weight='balanced'** vs RandomOverSampler.

Splitting

75/25 Train-Test split.

Models

6 runs across Logistic Regression, Random Forest, and XGBoost.

Model 1: Logistic Regression

Performance vs Base

METRIC	BASE	TUNED	CHANGE
Recall	0.421	0.720	+0.299
ROC-AUC	0.815	**0.815**	-
Precision	0.708	0.486	-0.222
F1	0.528	0.580	+0.052

Key Features

FEATURE	RANK
balance_drop_pct	#1
debit_to_balance	#2
no_last_txn	#3
prev_month_debit	#4

Model 2: Random Forest

Performance vs Base

METRIC	BASE	TUNED	CHANGE
Recall	0.592	0.703	+0.111
ROC-AUC	0.834	0.839	+0.005
Precision	0.638	0.512	-0.126

Key Features

FEATURE	RANK
balance_drop_pct	#1
current_balance	#2
Balance_change	#3
debit_to_balance	#4

Model 3: XGBoost Delivers

Performance vs Base

METRIC	BASE	TUNED	CHANGE
Recall	0.624	0.700	+0.076
ROC-AUC	0.816	0.842	+0.026
Precision	0.558	0.516	-0.042

Key Features

FEATURE	RANK
balance_drop_pct	#1
no_last_txn	#2
balance_change	#3
previous_month_debit	#4

XGBoost Is the Best Business Trade-off

All three models met the core targets:

Recall ≥ 0.70 and ROC-AUC ≥ 0.80.

Why XGBoost? It wins 4 of 5 metrics. While Logistic Regression has 0.020 higher recall, it results in 139 more wasted calls, a poor trade for the bank.

MODEL PERFORMANCE MATRIX

MODEL	REC.	PREC.	AUC
Baseline	0.000	0.000	0.500
LogReg	0.720	0.486	0.815
Random Forest	0.703	0.512	0.839
XGBoost	**0.700**	**0.516**	**0.842**

Efficiency Gain

- LogReg 947 churners, 1002 wasted calls.
- XGBoost identifies **920** churners with only **863** wasted calls.

Business Impact & Next Steps

The baseline accounts mainly non-churners; the final model finds about 7 in every 10.

1 **Score**

Score every customer monthly using the XGBoost pipeline.

2 **Rank**

Rank by churn probability to prioritize high-risk accounts.

3 **Contact**

Send top (highest-risk) accounts to the retention team.

4 **Retrain**

Track outcomes and retrain as new data arrives.

Limitations & Potential Improvements

Limitation	Impact
No source documentation	Provenance could not be formally verified
Snapshot, not full history	Only 2 months and 2 quarters per customer
No context on why	No complaints, competitor offers, or life events
Precision about 0.52	Half of flagged customers would not have churned
30% of churners missed	395 customers in the test set

POTENTIAL IMPROVEMENTS
Longer history, more months per customer.
Service records, complaints history, to capture why people leave
Estimate what a saved customer is worth, then optimize on revenue instead of a recall target

Thank You